EXERCISE 8

Reduction of Order for the Inhomogeneous Equation

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Reduction of Order • Integrating Factor • General Solution

STATEMENT

The so-called method of reduction of order can be used to find one solution of the inhomogeneous equation $L[y] = y'' + p(t)y' + q(t)y = g(t)$, provided one solution $y_1(t)$ of the corresponding homogeneous equation is known. Indeed, writing $y = v(t)y_1(t)$, show that $y(t)$ satisfies $L[y] = g$ if v is a solution of

$$y_1(t)v'' + \left[2y_1' + p(t)y_1\right]v' = g(t).$$

This is a first order linear equation for $v'(t)$. Solving this equation, integrating the result, and then multiplying by $y_1(t)$ leads to the general solution of the problem for $y(t)$.

SETUP AND HYPOTHESES

Hypotheses: p, q, g continuous on an interval I; y_1 is a known solution of $L[y_1] = 0$ with $y_1 \neq 0$ on I (so that we can divide by y_1). Unknown: $v(t)$, with $y = vy_1$.

FORMULATION

The idea is the same as in Topic 3, Sec. 7: promote the constant in Ay_1 to a function, $y = vy_1$. The product rule is used twice:

$$y' = v'y_1 + vy_1', \qquad y'' = v''y_1 + 2v'y_1' + vy_1''$$

CLASS NOTES AND COMPLEMENT [NOTES, P. 7]

The notes reduce the order of the homogeneous equation: with a known solution u, set $x = yu$; then $\big[u'' + pu' + qu\big]y + \big[2u' + pu\big]y' + uy'' = 0$, the first bracket is 0, and $z = y'$ satisfies the first-order equation $uz' + (2u' + pu)z = 0$. Here the notes' u is y_1 and their y is v; keeping the forcing on the right side is a complement (not in the class notes).

DERIVATION

Step 1: substitute in $L[y]$

$$L[y] = y'' + py' + qy = (v''y_1 + 2v'y_1' + vy_1'') + p\,(v'y_1 + vy_1') + qvy_1$$

Group the terms by v'', v' and v:

$$L[y] = y_1v'' + (2y_1' + py_1)\,v' + v\underbrace{(y_1'' + py_1' + qy_1)}_{L[y_1]=0}$$

The bracket multiplying v is $L[y_1]$, which is zero because y_1 solves the homogeneous equation; this is why the undifferentiated v disappears. Therefore

$$L[y] = y_1v'' + (2y_1' + py_1)\,v'$$

and $L[y] = g$ holds if and only if $y_1v'' + [2y_1' + py_1]v' = g$, which is the equation of the statement. ■

Step 2: solve the first-order equation for $w = v'$

Divide by $y_1 \neq 0$ and put $w = v'$:

$$w' + \left(\frac{2y_1'}{y_1} + p\right) w = \frac{g}{y_1}$$

This is a linear first-order equation (Topic 2). The integrating factor is $\mu = \exp\int\left(\frac{2y_1'}{y_1} + p\right)dt = y_1^2\, e^{P(t)}$, with $P(t) = \int p\,dt$ (because $\int \frac{2y_1'}{y_1}dt = 2\ln|y_1|$). The product rule gives $\frac{d}{dt}(\mu w) = \mu w' + \mu' w = \mu\left(w' + \left(\frac{2y_1'}{y_1} + p\right)w\right) = \mu\frac{g}{y_1}$:

$$\frac{d}{dt}\left(y_1^2 e^P w\right) = y_1 e^P g \quad\Longrightarrow\quad w(t) = \frac{1}{y_1^2 e^P}\left[\int y_1 e^P g\,dt + c_1\right]$$

Step 3: integrate for v and multiply by y_1

$$v(t) = \int w\,dt + c_2, \qquad y(t) = y_1(t)\,v(t) = y_1(t)\left\{c_2 + \int \frac{1}{y_1^2 e^P}\left[c_1 + \int y_1 e^P g\,dt\right]dt\right\}$$

This expression contains two arbitrary constants c_1, c_2. Setting $g = 0$ it reduces to the homogeneous solutions $y = c_2 y_1 + c_1 y_1 \int \frac{dt}{y_1^2 e^P}$ (the second solution of Robinson Exercise 17.7), and the part that does not contain c_1, c_2 is a particular solution of $L[y] = g$. The two homogeneous solutions y_1 and $y_2 = y_1\int\frac{dt}{y_1^2 e^P}$ are independent: by the product rule $y_2' = y_1'\int\frac{dt}{y_1^2e^P} + \frac{1}{y_1 e^P}$, so $W = y_1 y_2' - y_2 y_1' = e^{-P} \neq 0$ (the terms with the integral cancel). Hence complementary function plus particular integral, with two free constants, is the general solution (Robinson Sec. 14.1).

INTERPRETATION AND CHECKS

- The constant c_2 multiplies y_1 itself (the known solution) and c_1 multiplies $y_1\int\frac{dt}{y_1^2e^P}$, a second, independent homogeneous solution. Their Wronskian is $W = e^{-P}$, nowhere zero: consistent with Abel's identity $W = W_0\exp(-\int p)$.
- Check of the Wronskian: $y_2 = y_1\int\frac{dt}{y_1^2e^P}$ gives $y_2' = y_1'\int\frac{dt}{y_1^2e^P} + \frac{1}{y_1e^P}$, hence $W = y_1y_2' - y_2y_1' = \frac{1}{e^P}$ (the $y_1y_1'\int\ldots$ terms cancel).
- Compared with variation of constants, reduction of order needs only **one** homogeneous solution, at the price of two nested integrations.
- The method applies on any interval where $y_1 \neq 0$; where y_1 vanishes the division is not allowed and the intervals must be treated separately.